

02 – Keys and Normalization

The key is the key, let's be normal

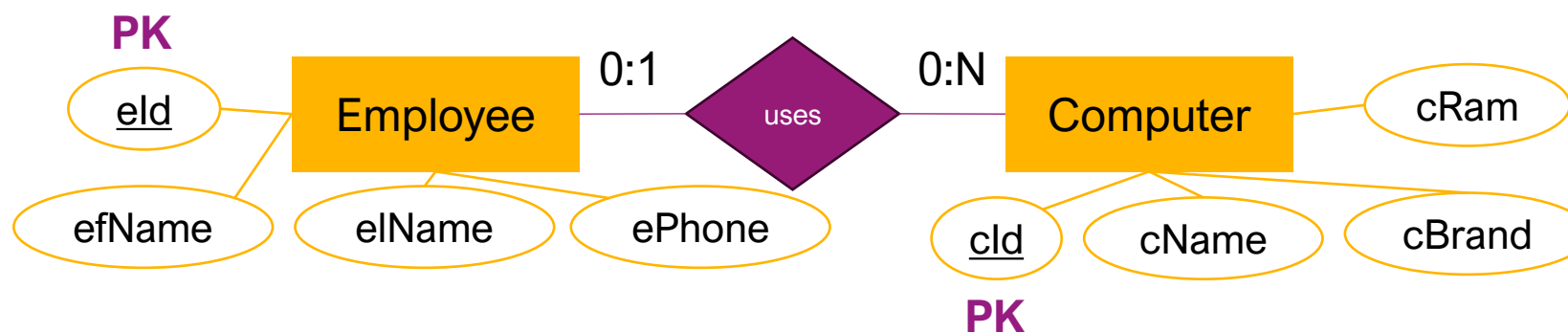
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OUTLINE

- 1) The key is the key
 - 1.1) The Primary Key (PK)
 - 1.2) The Foreign Key (FK)
 - 1.3) Other keys
- 2) Normalization
- 3) Anomalies
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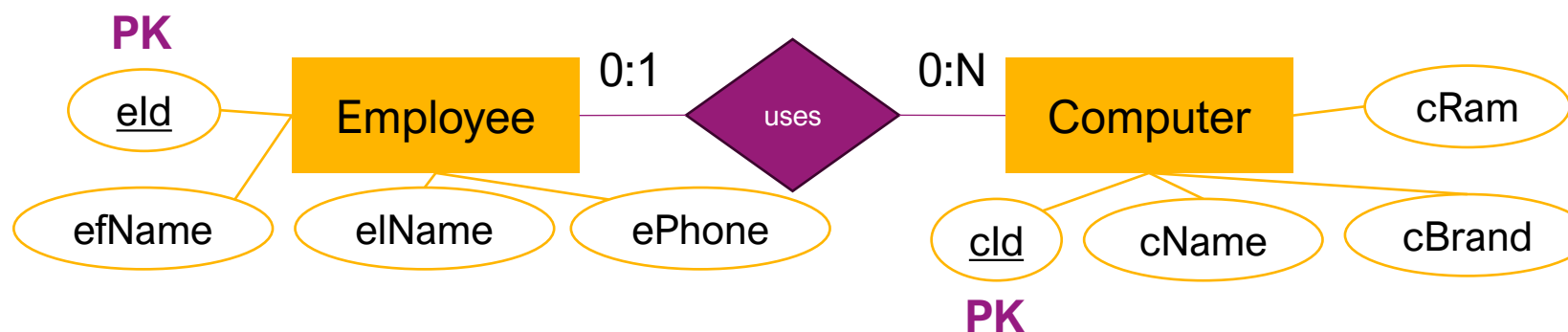
1.1) THE PRIMARY KEY

- In every **table**, we need a **PRIMARY KEY** that will uniquely define a tuple in this table.
- It is a fundamental concept in relational databases, you need **one PRIMARY KEY (PK) in each table**.

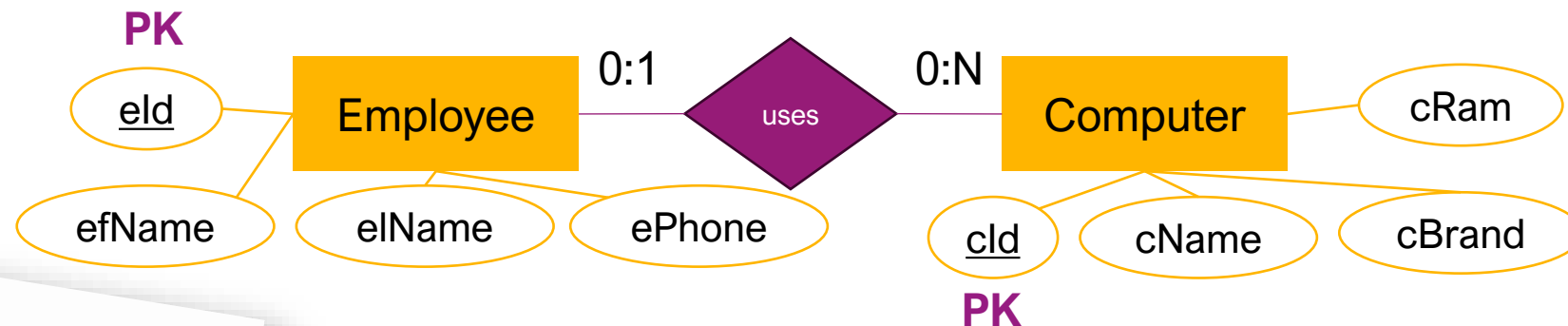


1.2) THE FOREIGN KEY

- There is no **FOREIGN KEY** on an Entity-Relationship Diagram (ERD)!
- The **FOREIGN KEY** comes from the rules that you apply to transform your conceptual (ERD) model into a logical model (tables):



1.2) THE FOREIGN KEY



RULES:

- 1) Any **entity** becomes a **table**.
- 2) Look at the **cardinalities (max only)**:
 - a. a **one-to-many** cardinality makes the **PK** on the 1 side become a **FK** into the table on the * side.
 - b. a **many-to-many** relation becomes a **full table** with **PK** being the two **FK** of the tables in the relation.

Computer

<u>cld</u>	cName	cBrand	cRam	eld
33	laptopLJ	HP	16GB	123
124	serverCSI	IBM	64GB	NULL
27	laptopMP	HP	24GB	241
185	desktop3 3	HP	16GB	241

Primary Key
PK

Foreign Key
FK

Employee

<u>eld</u>	efName	eName	ePhone
123	Jérôme	LANDRÉ	0708091011
241	Mira	POP	0607080910
324	Jasmin	JAKUPOVIC	0606060505

Primary Key
PK

1.3) OTHER KEYS

Super Key (SK): one or more attributes (columns) that can uniquely identify a tuple (record)

Candidate Key (CK): minimal set of attributes that can uniquely identify a tuple

Primary Key (PK): There can be more than one candidate key in relation out of which one **must** be chosen as the primary key

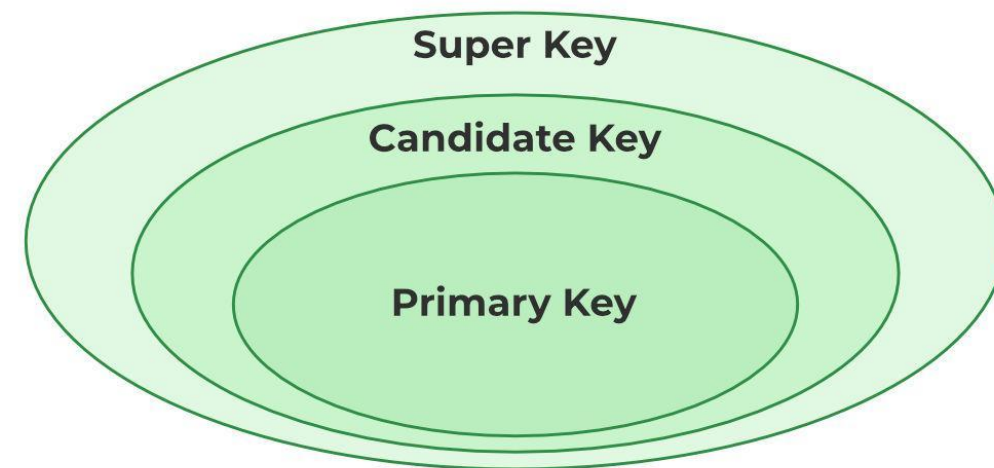


Table STUDENT

STUD_NO	SNAME	ADDRESS	PHONE
1	Shyam	Delhi	123456789
2	Rakesh	Kolkata	223365796
3	Suraj	Delhi	175468965

- 1) SK: STUD_NO+PHONE // STUD_NO+SNAME
- 2) CK: STUD_NO // PHONE
- 3) PK: STUD_NO

1.3) OTHER KEYS

Alternate Key (AK) or Secondary Key: any candidate key in a table that is **not** chosen as the **primary key**

Composite Key (CompK): a key made of one or more attributes (columns)

Foreign Key (FK): an attribute in one table that refers to the **primary key** in another table

2) NORMALIZATION

Chapter 12 Normalization

ADRIENNE WATT

Normalization should be part of the database design process. However, it is difficult to separate the normalization process from the ER modelling process so the two techniques should be used concurrently.

Use an entity relation diagram (ERD) to provide the big picture, or macro view, of an organization's data requirements and operations. This is created through an iterative process that involves identifying relevant entities, their attributes and their relationships.

Normalization procedure focuses on characteristics of specific entities and represents the micro view of entities within the ERD.

2) NORMALIZATION

What Is Normalization?

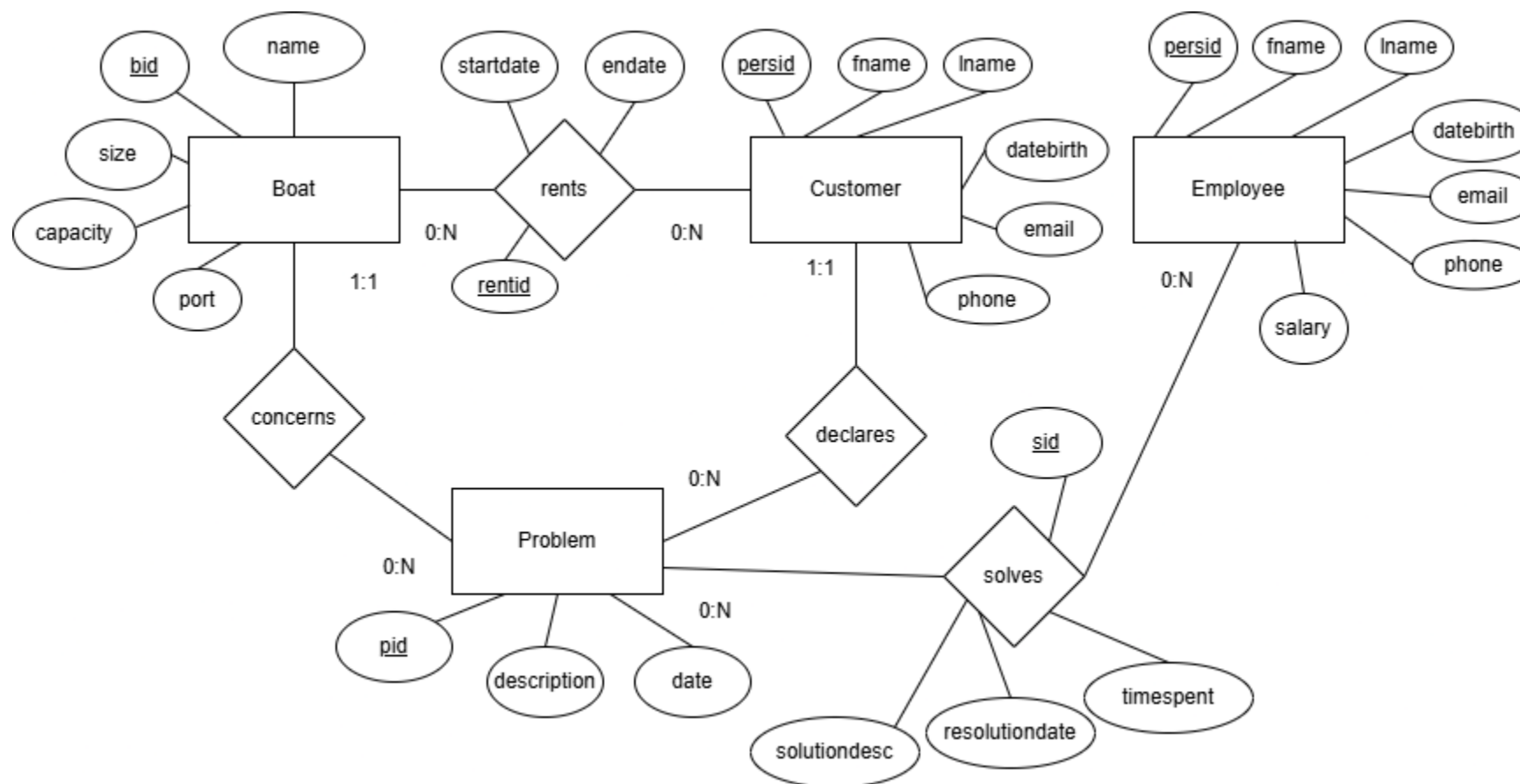
Normalization is the branch of relational theory that provides design insights. It is the process of determining how much redundancy exists in a table. The goals of normalization are to:

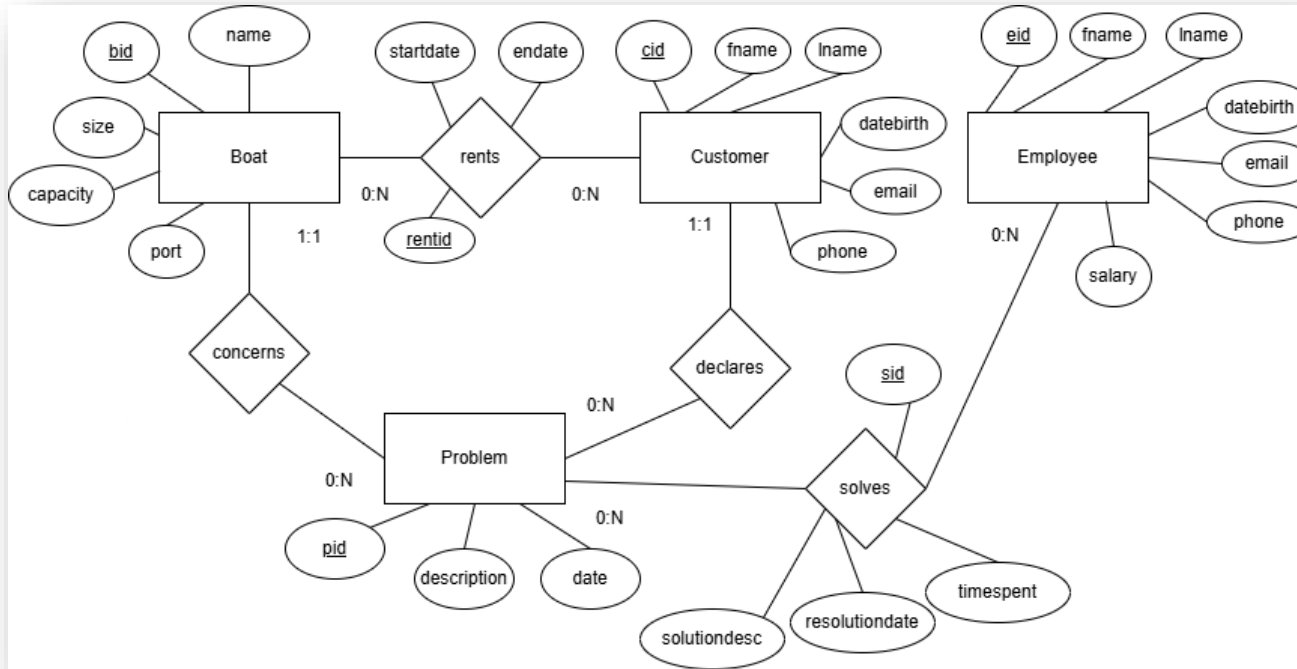
- Be able to characterize the level of redundancy in a relational schema
- Provide mechanisms for transforming schemas in order to remove redundancy

Normalization theory draws heavily on the theory of functional dependencies. Normalization theory defines six normal forms (NF). Each normal form involves a set of dependency properties that a schema must satisfy and each normal form gives guarantees about the presence and/or absence of update anomalies. This means that higher normal forms have less redundancy, and as a result, fewer update problems.

3) ANOMALIES

Example (from workshop 1): we have Customer and Employee in two different **entities**, we have **redundancy of information** if an employee is also a customer.





INSERTION ANOMALY

Employee

123 | Jérôme | Landré | 1972-01-01 | jerome.landre@ju.se | 0707070707 | 999 999 999

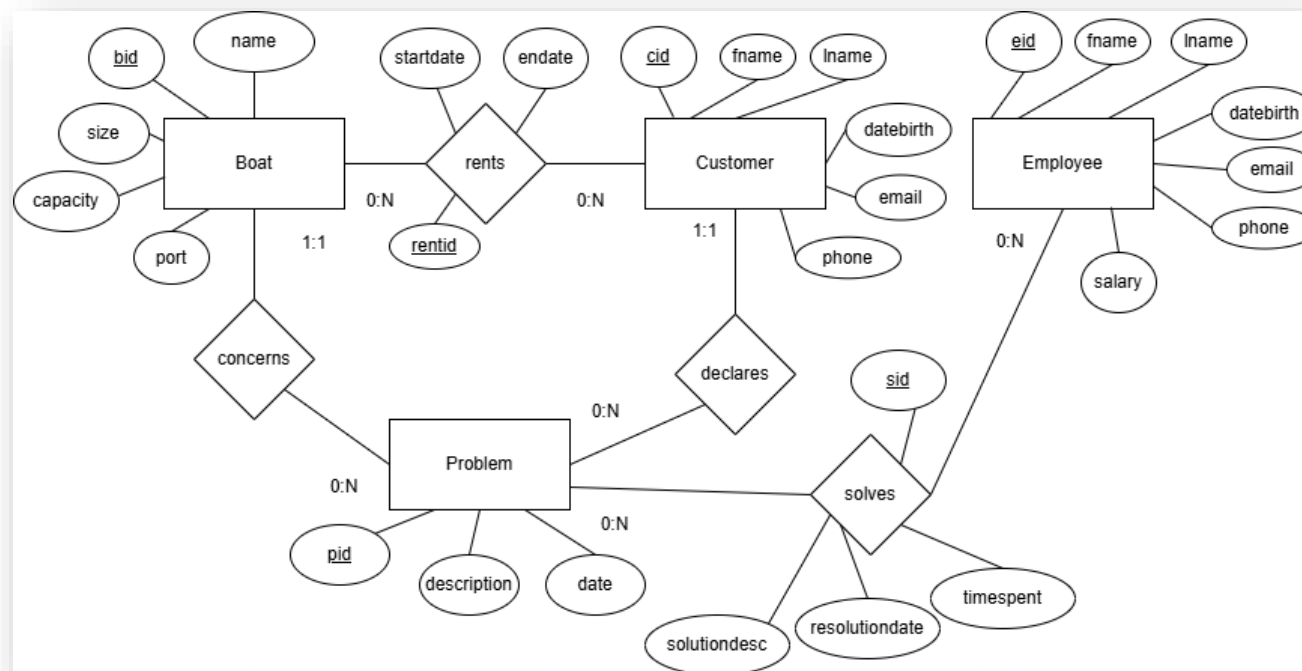
Customer

579 | Jérôme | Landér | 1972-01-01 | jerome.landre@ju.se | 0707070707

If Jérôme is also a Customer, I can insert the (wrong) information in the other table...

Firstly, it is an **insertion anomaly**.

Secondly, I don't know that these two "Jérôme"s are the same person!



UPDATE ANOMALY

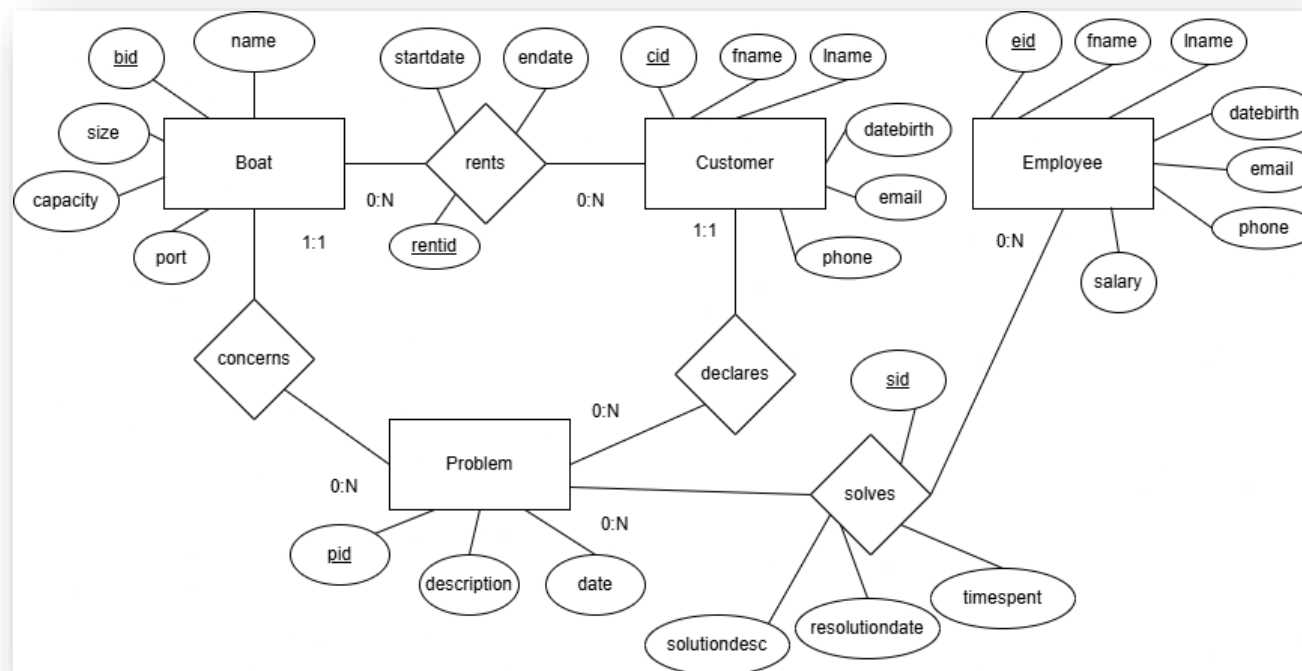
Employee

123 | Jérôme | Landré | 1972-01-01 | jerome.landre@ju.se | 0707070707 | 999 999 999

Customer

579 | Jérôme | Landér | 1972-01-01 | jerome.landre@ju.se | 0607080910

If I update the phone number because Jérôme changed his number, I must do it twice!
 Firstly, it is an **update anomaly**.
 Secondly, I still don't know that these two "Jérôme"s are the same person!



DELETION ANOMALY

Employee

123 | Jérôme | Landré | 1972-01-01 | jerome.landre@ju.se | 0707070707 | 999 999 999

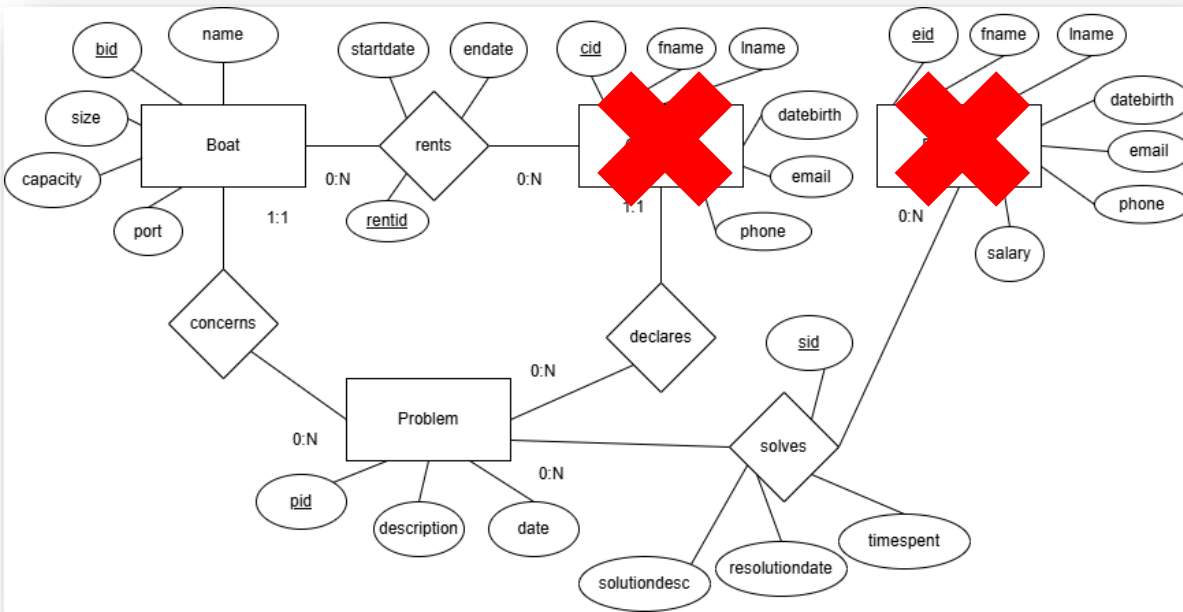
Customer

~~579 | Jérôme | Landré | 1972-01-01 | jerome.landre@ju.se | 0607080910~~

Problem

3028 | Broken door. | 2026-03-27 | 579

If I delete Customer 579, the problem will not have any Customer associated!
It is a **deletion anomaly**.

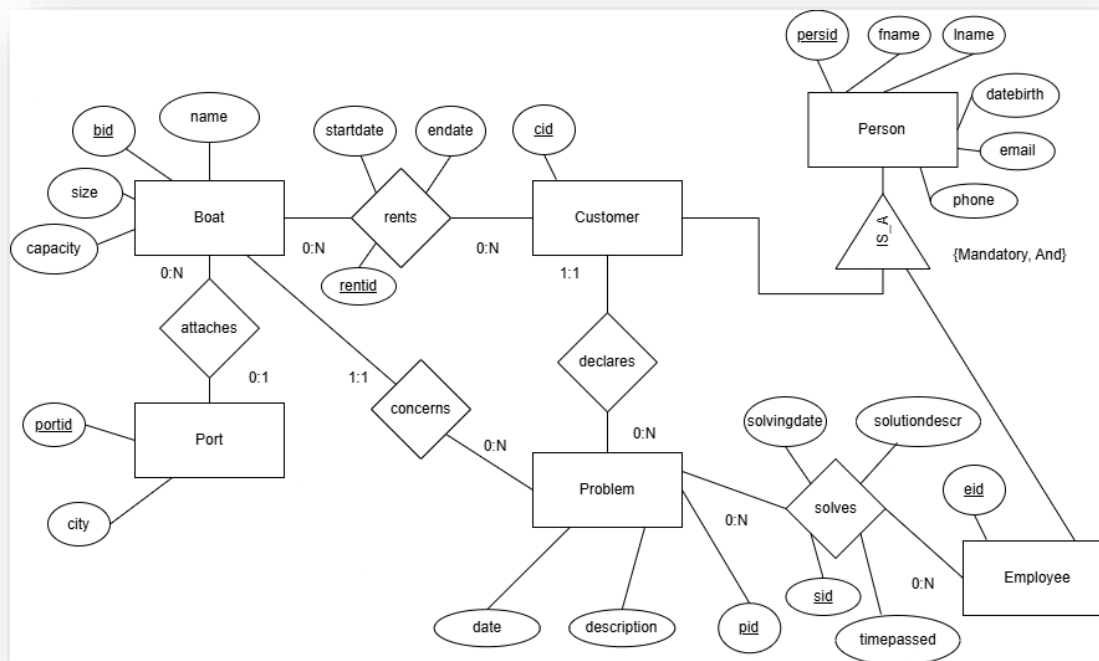


What is the problem?

Data redundancy: multiple time the same data at different locations in the database!

What is the solution?

Avoid data redundancy: each piece of **data** is stored in **only one place** (or data is never duplicated).



4) NORMALIZATION IN PRACTICE

What is normalization?

Normalization is a technique used with database design which serves the purpose of creating tables with reduced:

redundancy and
dependency of data

Why be normal?

It's not *cool* to be messy when it comes to databases. You want to be able to search for everything.

It's a process of making "big" tables "smaller" and splitting the data around for the purpose of reusing it in other tables.

Normalization makes searching much, much easier. It also helps with a better visual representation of the database. Good database design means **enough independent** tables

4) NORMALIZATION IN PRACTICE

Sounds complicated?

Not so easy, so...

... keep it simple: follow the rules!

4) NORMALIZATION IN PRACTICE

Normal Forms

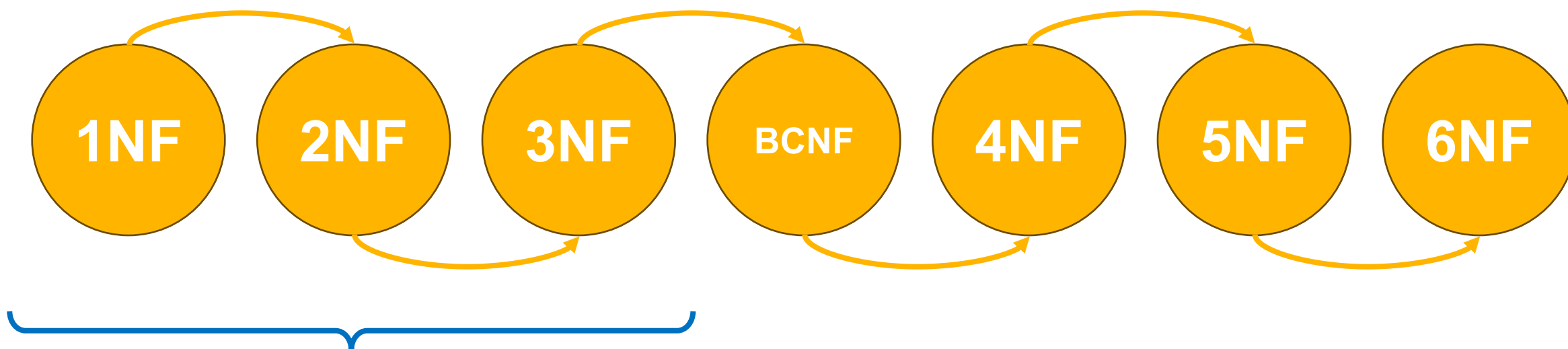
All the tables in any database can be in one of the normal forms we will discuss next. Ideally we only want minimal redundancy for PK to FK. Everything else should be derived from other tables. There are six normal forms, but we will only look at the first four, which are:

- First normal form (1NF)
- Second normal form (2NF)
- Third normal form (3NF)
- Boyce-Codd normal form (BCNF)

BCNF is rarely used.

4) NORMALIZATION IN PRACTICE

Normal Forms:



The 3 first normal forms: **1NF**, **2NF** and **3NF**
Can **ALWAYS** be applied to any database!

4) NORMALIZATION IN PRACTICE

Smart Table Design

When developing databases, it's very important to think about normalization and the normal forms.

But, even without knowing the normalization rules you ought to strive for making your database tables "smart".

There is one rule with *smart table design* which is, **group** and **re-use**.

Grouping means that you identify all potential columns that could be stored in separate standalone tables.

Re-using is a natural outcome when grouping. You want to be able to use data from one table in one or many other tables.

Go with **Primary Keys** instead of **Composite Keys**.

4) NORMALIZATION IN PRACTICE

Normalization, or **NFs**, are based on simple rules and the study of **functional dependencies** (**FDs**) between the attributes in the relations (tables).

The process of normalization is taking an existing schema through a series of **'forms'** or **'tests'**.

It is an **iterative** (progressive) process: once it fulfills one form you can proceed to the next one.

TIPS AND TRICKS

BE POSITIVE!

« Tu l'sais, dans la vie, on s'est tous plantés
C'est vrai, combien d'fois on a dû se relever ?
Personne n'est parfait, on est tous sortis du chemin
Tu sais, dans la vie, ça va, ça vient... »

"You know, in life, we've all messed up.
It's true, how many times have we had to get back up?
Nobody's perfect, we've all strayed from the path.
You know, in life, things come and go..."

"Du vet, i livet har vi alla gjort fel.
Det är sant, hur många gånger har vi varit tvungna att resa oss upp igen?
Ingen är perfekt, vi har alla kommit vilse från vägen.
Du vet, i livet kommer och går saker..."

Vitaa et Slimane, <https://www.youtube.com/watch?v=Wz86pFtHZqI>

5) FUNCTIONAL DEPENDENCIES

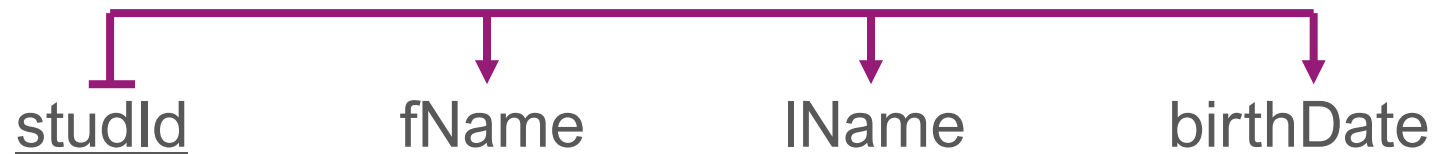
Chapter 11 Functional Dependencies

ADRIENNE WATT

A *functional dependency* (FD) is a relationship between two attributes, typically between the PK and other non-key attributes within a table. For any relation R, attribute Y is functionally dependent on attribute X (usually the PK), if for every valid instance of X, that value of X uniquely determines the value of Y. This relationship is indicated by the representation below :

$X \text{ ----} \rightarrow Y$

The left side of the above FD diagram is called the *determinant*, and the right side is the *dependent*. Here are a few examples.



5) FUNCTIONAL DEPENDENCIES

In the first example, below, SIN determines Name, Address and Birthdate. Given SIN, we can determine any of the other attributes within the table.

SIN ----> Name, Address, Birthdate

For the second example, SIN and Course determine the date completed (DateCompleted). This must also work for a composite PK.

SIN, Course ----> DateCompleted

The third example indicates that ISBN determines Title.

ISBN ----> Title

6) NORMAL FORMS: 1NF, 2NF, 3NF, ...

First Normal Form (1NF)

In the *first normal form*, only single values are permitted at the intersection of each row and column; hence, there are no repeating groups.

4) NORMAL FORMS: 1NF, 2NF, 3NF, ...

Second Normal Form (2NF)

For the *second normal form*, the relation must first be in 1NF. The relation is automatically in 2NF if, and only if, the PK comprises a single attribute.

4) NORMAL FORMS: 1NF, 2NF, 3NF, ...

Third Normal Form (3NF)

To be in *third normal form*, the relation must be in second normal form. Also all transitive dependencies must be removed; a non-key attribute may not be functionally dependent on another non-key attribute.

First Normal Form (1NF)

In the *first normal form*, only single values are permitted at the intersection of each row and column; hence, there are no repeating groups.

Second Normal Form (2NF)

For the *second normal form*, the relation must first be in 1NF. The relation is automatically in 2NF if, and only if, the PK comprises a single attribute.

Third Normal Form (3NF)

To be in *third normal form*, the relation must be in second normal form. Also all transitive dependencies must be removed; a non-key attribute may not be functionally dependent on another non-key attribute.

4) NORMAL FORMS: 1NF, 2NF, 3NF, ...

Here comes the **Normal Form rules**:

- **1NF**: Each column in a table must contain atomic (indivisible) values, and each row must be unique.

This means that no column should contain multiple values, and each cell should hold a single value. Additionally, the table must be two-dimensional with rows and columns, and each column must have a unique name.

- **2NF**: A table must already be in 1NF and must not have partial dependencies.

This means that every non-key attribute must be fully dependent on the entire primary key.

If the primary key is a composite key (consisting of multiple columns), then all non-key attributes must depend on the entire primary key, not just a part of it.

- **3NF**: A table must already be in 2NF and must not have transitive dependencies.

This means that every non-key attribute must be directly dependent on the primary key and not on any other non-key attribute.

7) EXAMPLES

7.1) EXAMPLE 1

In this example you have a spreadsheet to keep track of your studies. There, you have a list of courses you are taking, responsible teachers, credits, topics, examinations... You also want to know where each teacher sits so you can go and visit them if required. It might look like this:

Course	Topic	Teacher	Office	Phone	Title	HP	Examination
Programming	Computer Science	Anders Carlsson	House E, Room 2414	0707070707	Lic.	7.5	exam, labs
Databases	CS	Jasmin Jakupovic	House E, Room 3421	0607080910	MSc.	6	exam, project, labs
Algebra	Math	Fredrik Ek	House K, Room 45	0612345678	Dr.	9	exam
Software Engineering	Computer Science	Jasmin Jakupovic	House E, Room 3421	0607080910	MSc.	7.5	exam, project
Statistics	Maths	Fredrik Ek	House K, Room 45	0612345678	Dr.	7.5	exam

Course	Topic	Teacher	Office	Phone	Title	HP	Examination
Programming	Computer Science	Anders Carlsson	House E, Room 2414	0707070707	Lic.	7.5	exam, labs
Databases	CS	Jasmin Jakupovic	House E, Room 3421	0607080910	MSc.	6	exam, project, labs
Algebra	Math	Fredrik Ek	House K, Room 45	0612345678	Dr.	9	exam
Software Engineering	Computer Science	Jasmin Jakupovic	House E, Room 3421	0607080910	MSc.	7.5	exam, project
Statistics	Maths	Fredrik Ek	House K, Room 45	0612345678	Dr.	7.5	exam

Which poor limitations can we identify?

1. Inconsistent naming
2. Name of teacher (must be firstname & lastname)
3. Office (address) in one column
4. Title directly dependant from teacher
5. Multiple examinations in one column

1NF Rules:

- Each table cell should contain a **single** value
- Each record needs to be **unique**

Course	Topic	Teacher	Office	Phone	Title	HP	Examination
Programming	Computer Science	Anders Carlsson	House E, Room 2414	0707070707	Lic.	7.5	exam, labs
Databases	CS	Jasmin Jakupovic	House E, Room 3421	0607080910	MSc.	6	exam, project, labs
Algebra	Math	Fredrik Ek	House K, Room 45	0612345678	Dr.	9	exam
Software Engineering	Computer Science	Jasmin Jakupovic	House E, Room 3421	0607080910	MSc.	7.5	exam, project
Statistics	Maths	Fredrik Ek	House K, Room 45	0612345678	Dr.	7.5	exam

Course	Topic	fName	IName	House	Room	Phone	Title	HP	Examination
Programming	Computer Science	Anders	Carlsson	E	2414	0707070707	Lic.	7.5	exam
Programming	Computer Science	Anders	Carlsson	E	2414	0707070707	Lic.	7.5	labs
Databases	Computer Science	Jasmin	Jakupovic	E	3421	0607080910	MSc.	6	exam
Databases	Computer Science	Jasmin	Jakupovic	E	3421	0607080910	MSc.	6	project
Databases	Computer Science	Jasmin	Jakupovic	E	3421	0607080910	MSc.	6	labs
Algebra	Math	Fredrik	Ek	K	45	0612345678	Dr.	9	exam
Software Engineering	Computer Science	Jasmin	Jakupovic	E	3421	0798765432	MSc.	7.5	exam
Software Engineering	Computer Science	Jasmin	Jakupovic	E	3421	0798765432	MSc.	7.5	project
Statistics	Maths	Fredrik	Ek	K	45	0706050403	Dr.	7.5	exam

1NF

Course	Topic	fName	IName	House	Room	Phone	Title	HP	Examination
Programming	Computer Science	Anders	Carlsson	E	2414	0707070707	Lic.	7.5	exam
Programming	Computer Science	Anders	Carlsson	E	2414	0707070707	Lic.	7.5	labs
Databases	Computer Science	Jasmin	Jakupovic	E	3421	0607080910	MSc.	6	exam
Databases	Computer Science	Jasmin	Jakupovic	E	3421	0607080910	MSc.	6	project
Databases	Computer Science	Jasmin	Jakupovic	E	3421	0607080910	MSc.	6	labs
Algebra	Math	Fredrik	Ek	K	45	0706050403	Dr.	9	exam
Software Engineering	Computer Science	Jasmin	Jakupovic	E	3421	0798765432	MSc.	7.5	exam
Software Engineering	Computer Science	Jasmin	Jakupovic	E	3421	0798765432	MSc.	7.5	project
Statistics	Maths	Fredrik	Ek	K	45	0706050403	Dr.	7.5	exam

2NF Rules:

- Be in 1NF
- Single Column Primary Key

Cid	Course	Topic	fName	IName	House	Room	Phone	Title	HP	Examination
1	Programming	Computer Science	Anders	Carlsson	E	2414	0707070707	Lic.	7.5	exam
1	Programming	Computer Science	Anders	Carlsson	E	2414	0707070707	Lic.	7.5	labs
2	Databases	Computer Science	Jasmin	Jakupovic	E	3421	0607080910	MSc.	6	exam
2	Databases	Computer Science	Jasmin	Jakupovic	E	3421	0607080910	MSc.	6	project
2	Databases	Computer Science	Jasmin	Jakupovic	E	3421	0607080910	MSc.	6	labs
3	Algebra	Math	Fredrik	Ek	K	45	0706050403	Dr.	9	exam
4	Software Engineering	Computer Science	Jasmin	Jakupovic	E	3421	0798765432	MSc.	7.5	exam
4	Software Engineering	Computer Science	Jasmin	Jakupovic	E	3421	0798765432	MSc.	7.5	project
5	Statistics	Maths	Fredrik	Ek	K	45	0706050403	Dr.	7.5	exam

2NF Rules:

- Be in 1NF
- Single Column Primary Key

Cid	Course	Topic	fName	IName	House	Room	Phone	Title	HP	Examination
1	Programming	Computer Science	Anders	Carlsson	E	2414	0707070707	Lic.	7.5	exam
1	Programming	Computer Science	Anders	Carlsson	E	2414	0707070707	Lic.	7.5	labs
2	Databases	Computer Science	Jasmin	Jakupovic	E	3421	0607080910	MSc.	6	exam
2	Databases	Computer Science	Jasmin	Jakupovic	E	3421	0607080910	MSc.	6	project
2	Databases	Computer Science	Jasmin	Jakupovic	E	3421	0607080910	MSc.	6	labs
3	Algebra	Math	Fredrik	Ek	K	45	0706050403	Dr.	9	exam
4	Software Engineering	Computer Science	Jasmin	Jakupovic	E	3421	0798765432	MSc.	7.5	exam
4	Software Engineering	Computer Science	Jasmin	Jakupovic	E	3421	0798765432	MSc.	7.5	project
5	Statistics	Maths	Fredrik	Ek	K	45	0706050403	Dr.	7.5	exam

2NF Rules:

- Be in 1NF
- Single Column Primary Key

Cid	Course	Topic	Examination
1	Programming	Computer Science	exam
1	Programming	Computer Science	labs
2	Databases	Computer Science	exam
2	Databases	Computer Science	project
2	Databases	Computer Science	labs
3	Algebra	Math	exam
4	Software Engineering	Computer Science	exam
4	Software Engineering	Computer Science	project
5	Statistics	Maths	exam

We put a number in front of all different courses: Cid. For each course, there should be only one identifier obviously.

We have a problem there: we must have a different Cid for each row. But the course id appears in multiple rows, due to the different examinations in several courses.

There is only one way to solve the problem: remove the column examination and put it in another table!

<u>Cid</u>	Course	Topic	fName	lName	House	Room	Phone	Title	HP
1	Programming	Computer Science	Anders	Carlsson	E	2414	0707070707	Lic.	7.5
2	Databases	Computer Science	Jasmin	Jakupovic	E	3421	0607080910	MSc.	6
3	Algebra	Math	Fredrik	Ek	K	45	0612345678	Dr.	9
4	Software Engineering	Computer Science	Jasmin	Jakupovic	E	3421	0798765432	MSc.	7.5
5	Statistics	Maths	Fredrik	Ek	K	45	0706050403	Dr.	7.5

- Cid is the **PRIMARY KEY (PK)**
- It is **UNIQUE** in the table

- Cid is a FOREIGN KEY (FK)
- Cid+Examination is the PK
- Problem: **NOT UNIQUE** in the table

2NF Rules:

- Be in 1NF
- Single Column Primary Key

<u>Cid</u>	<u>Examination</u>
1	exam
1	labs
2	exam
2	labs
2	project
3	exam
4	exam
4	project
5	exam

Cid	Course	Topic	fName	lName	House	Room	Phone	Title	HP
1	Programming	Computer Science	Anders	Carlsson	E	2414	0707070707	Lic.	7.5
2	Databases	Computer Science	Jasmin	Jakupovic	E	3421	0607080910	MSc.	6
3	Algebra	Math	Fredrik	Ek	K	45	0612345678	Dr.	9
4	Software Engineering	Computer Science	Jasmin	Jakupovic	E	3421	0798765432	MSc.	7.5
5	Statistics	Maths	Fredrik	Ek	K	45	0706050403	Dr.	7.5

- Cid is the PRIMARY KEY (PK)
- It is **UNIQUE** in the table

- CoursExam is the PK
- It is **UNIQUE** in the table

CoursExam	Cid	Examination
1	1	exam
2	1	labs
3	2	exam
4	2	labs
5	2	project
6	3	exam
7	4	exam
8	4	project
9	5	exam

2NF Rules:

- Be in 1NF
- Single Column Primary Key

Let's go for **3NF**

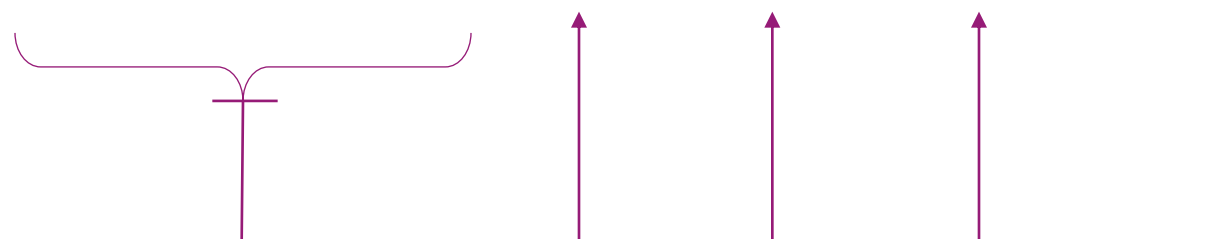
7.1) EXAMPLE 1

3NF Rules:

- Be in 2NF
- Has no transitive functional dependencies

A transitive functional dependency is when changing a non-key (ordinary) column, might cause any other column to change. Imagine changing Column A which then immediately impacts the value (indirectly) in Column B.

Cid	Course	Topic	fName	IName	House	Room	Phone	Title	HP
1	Programming	Computer Science	Anders	Carlsson	E	2414	0707070707	Lic.	7.5
2	Databases	Computer Science	Jasmin	Jakupovic	E	3421	0607080910	MSc.	6
3	Algebra	Math	Fredrik	Ek	K	45	0612345678	Dr.	9
4	Software Engineering	Computer Science	Jasmin	Jakupovic	E	3421	0798765432	MSc.	7.5
5	Statistics	Maths	Fredrik	Ek	K	45	0706050403	Dr.	7.5



3NF Rules:

- Be in 2NF
- Has no transitive functional dependencies

<u>Cid</u>	Course	Topic	Tid	HP
1	Programming	Computer Science	1	7.5
2	Databases	Computer Science	2	6
3	Algebra	Math	3	9
4	Software Engineering	Computer Science	2	7.5
5	Statistics	Maths	3	7.5

3NF Rules:

- Be in 2NF
- Has no transitive functional dependencies

<u>Tid</u>	fName	lName	House	Room	Phone	Title
1	Anders	Carlsson	E	2414	0707070707	Lic.
2	Jasmin	Jakupovic	E	3421	0607080910	MSc.
3	Fredrik	Ek	K	45	0612345678	Dr.

<u>Cid</u>	Course	Topic	HP	Tid
1	Programming	Computer Science	7.5	1
2	Databases	Computer Science	6	2
3	Algebra	Math	9	3
4	Software Engineering	Computer Science	7.5	2
5	Statistics	Maths	7.5	3

3NF Rules:

- Be in 2NF
- Has no transitive functional dependencies

<u>Tid</u>	fName	IName	House	Room	Phone	Title
1	Anders	Carlsson	E	2414	0707070707	Lic.
2	Jasmin	Jakupovic	E	3421	0607080910	MSc.
3	Fredrik	Ek	K	45	0612345678	Dr.

<u>Cid</u>	Course	Topic	HP	Tid
1	Programming	Computer Science	7.5	1
2	Databases	Computer Science	6	2
3	Algebra	Math	9	3
4	Software Engineering	Computer Science	7.5	2
5	Statistics	Maths	7.5	3

<u>CoursExam</u>	Cid	Examination
1	1	exam
2	1	labs
3	2	exam
4	2	labs
5	2	project
6	3	exam
7	4	exam
8	4	project
9	5	exam

3NF Rules:

- Be in 2NF
- Has no transitive functional dependencies

<u>Tid</u>	fName	IName	House	Room	Phone	Title
1	Anders	Carlsson	E	2414	0707070707	Lic.
2	Jasmin	Jakupovic	E	3421	0607080910	MSc.
3	Fredrik	Ek	K	45	0612345678	Dr.

<u>Cid</u>	Course	Topic	HP	Tid
1	Programming	Computer Science	7.5	1
2	Databases	Computer Science	6	2
3	Algebra	Math	9	3
4	Software Engineering	Computer Science	7.5	2
5	Statistics	Maths	7.5	3

<u>CoursExam</u>	Cid	Examination
1	1	exam
2	1	labs
3	2	exam
4	2	labs
5	2	project
6	3	exam
7	4	exam
8	4	project
9	5	exam

<u>TitleId</u>	Titlename
1	Lic.
2	Msc.
3	Dr.

3NF Rules:

- Be in 2NF
- Has no transitive functional dependencies

<u>Tid</u>	fName	lName	House	Room	Phone	TitleId
1	Anders	Carlsson	E	2414	0707070707	1
2	Jasmin	Jakupovic	E	3421	0607080910	2
3	Fredrik	Ek	K	45	0612345678	3

7.1) EXAMPLE 1

Boyce-Codd Normal Form:

- Slightly stronger 3NF
- Often referred to as 3.5NF
- Addresses certain anomalies not dealt in 3NF
- Some redundancies may exist
- Functional dependency has been removed
- 3NF usually always meets BCNF
- Some exceptions might happen

7.1) EXAMPLE 1

4NF:

Table instances must not contain more than one independent and multi-valued data describing the relevant entity.

5NF:

Can not be decomposed from the 4NF into any number of smaller tables without any data loss.

6NF:

Not yet standardized!

Decomposition of relation variables to irreducible Components.

7.2) EXAMPLE 2: "FUNCTIONAL DEPENDENCIES"

CourseCode	Course	Topic	Teacher	Phone	Title	Hp
TIPS20	Programming	Computer Science	Anders Carstensen	1234	Lic.	7,5
TDRK17	Databases	Computer Science	Jasmin Jakupovic	4567	MSc.	6
TALG88	Algebra	Math	Fredrik Ek	89755	Dr.	9
TSEN05	Software Eng.	Computer Science	Jasmin Jakupovic	4567	MSc.	7,5
TSTA11	Statistics	Math	Fredrik Ek	89755	Dr.	7,5

Non-primary attributes

Candidate Keys

$R = (\text{CourseCode}, \text{Course}, \text{Topic}, \text{Teacher}, \text{Phone}, \text{Title}, \text{Hp})$

$F = (\underline{\text{CourseCode}} \rightarrow \text{Course}, \underline{\text{CourseCode}} \rightarrow \text{Topic}, \underline{\text{Teacher}} \rightarrow \text{Phone}, \underline{\text{Teacher}} \rightarrow \text{Title}, \underline{\text{CourseCode}} \rightarrow \text{Hp})$

Let's bring it to 2NF and 3NF

"Given the value of CourseCode we know Course"

7.2) EXAMPLE 2: "FUNCTIONAL DEPENDENCIES"

2NF Rules:

Be in 1NF and Single Column Primary Key

$R = (\text{CourseCode}, \text{Course}, \text{Topic}, \text{Teacher}, \text{Phone}, \text{Title}, \text{Hp})$

$F = (\underline{\text{CourseCode}} \rightarrow \text{Course}, \underline{\text{CourseCode}} \rightarrow \text{Topic}, \underline{\text{Teacher}} \rightarrow \text{Phone}, \underline{\text{Teacher}} \rightarrow \text{Title}, \underline{\text{CourseCode}} \rightarrow \text{Hp})$

Step 1 - Single Column Primary Key

$R_1 = (\text{CourseCode}, \text{Course}, \text{Topic}, \text{Hp})$

$F_1 = (\underline{\text{CourseCode}} \rightarrow \text{Course}, \underline{\text{CourseCode}} \rightarrow \text{Topic}, \underline{\text{CourseCode}} \rightarrow \text{Hp})$

$R_2 = (\text{Teacher}, \text{Phone}, \text{Title})$

$F_2 = (\underline{\text{Teacher}} \rightarrow \text{Phone}, \underline{\text{Teacher}} \rightarrow \text{Title})$

Step 2 - What remains from the CK's?

$R_0 = (\text{CourseCode}, \text{Teacher})$

$F_0 = (\emptyset)$



Empty, no dependencies

7.2) EXAMPLE 2: "FUNCTIONAL DEPENDENCIES"

Remember "Transitive Functional Dependencies"?

A transitive functional dependency is when changing a non-key (ordinary) column, might cause any other column to change. Imagine changing Column A which then immediately impacts the value (indirectly) in Column C.

R

A	B	C

$R = (A, B, C)$

Transitive
Functional Dependency

$A \twoheadrightarrow B$

$B \twoheadrightarrow C$

Functional Dependencies

7.2) EXAMPLE 2: "FUNCTIONAL DEPENDENCIES"

3NF Rules:

Be in 2NF and Has no transitive FD's

$R_1 = (\text{CourseCode}, \text{Course}, \text{Topic}, \text{Hp})$

$R_2 = (\text{Teacher}, \text{Phone}, \text{Title})$

$R_0 = (\text{CourseCode}, \text{Teacher})$

$F_1 = (\underline{\text{CourseCode}} \rightarrow \text{Course}, \underline{\text{CourseCode}} \rightarrow \text{Topic}, \underline{\text{CourseCode}} \rightarrow \text{Hp})$

$F_2 = (\underline{\text{Teacher}} \rightarrow \text{Phone}, \underline{\text{Teacher}} \rightarrow \text{Title})$

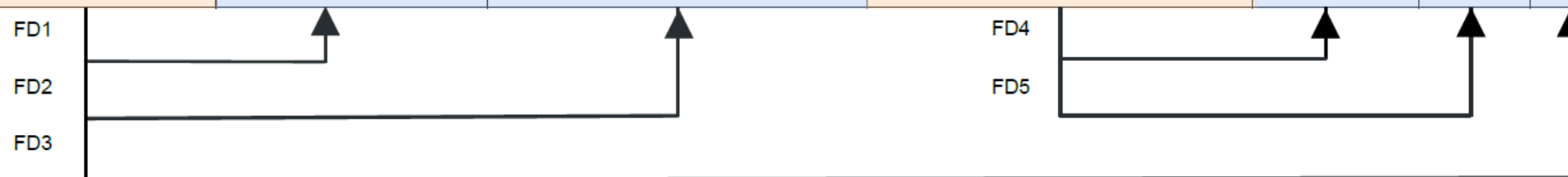
$F_0 = (\emptyset)$

There are no transitive functional dependencies so it is in 3NF!

7.2) EXAMPLE 2: "FUNCTIONAL DEPENDENCIES"

Diagrammatic Notation

CourseCode	Course	Topic	Teacher	Phone	Title	Hp
TIPS20	Programming	Computer Science	Anders Carstensen	1234	Lic.	7,5
TDRK17	Databases	Computer Science	Jasmin Jakupovic	4567	MSc.	6
TALG88	Algebra	Math	Fredrik Ek	89755	Dr.	9
TSEN05	Software Eng.	Computer Science	Jasmin Jakupovic	4567	MSc.	7,5
TSTA11	Statistics	Math	Fredrik Ek	89755	Dr.	7,5



$F = (\underline{\text{CourseCode}} \rightarrow \text{Course}, \underline{\text{CourseCode}} \rightarrow \text{Topic}, \underline{\text{Teacher}} \rightarrow \text{Phone}, \underline{\text{Teacher}} \rightarrow \text{Title}, \underline{\text{CourseCode}} \rightarrow \text{Hp})$

$R = (\text{CourseCode}, \text{Course}, \text{Topic}, \text{Teacher}, \text{Phone}, \text{Title}, \text{Hp})$

ANY QUESTIONS?